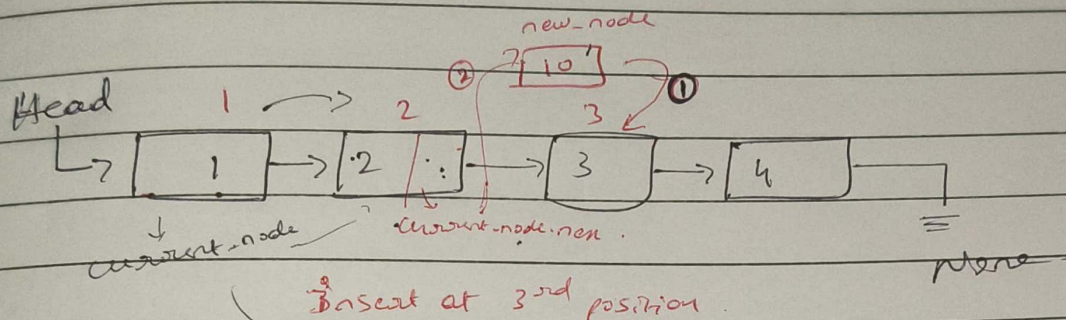


3 SLL in Python - Insert Node at Given position

Singly Linked List (SLL)

Step 1: Define Node:-

* Insert / Delete At Given position :-

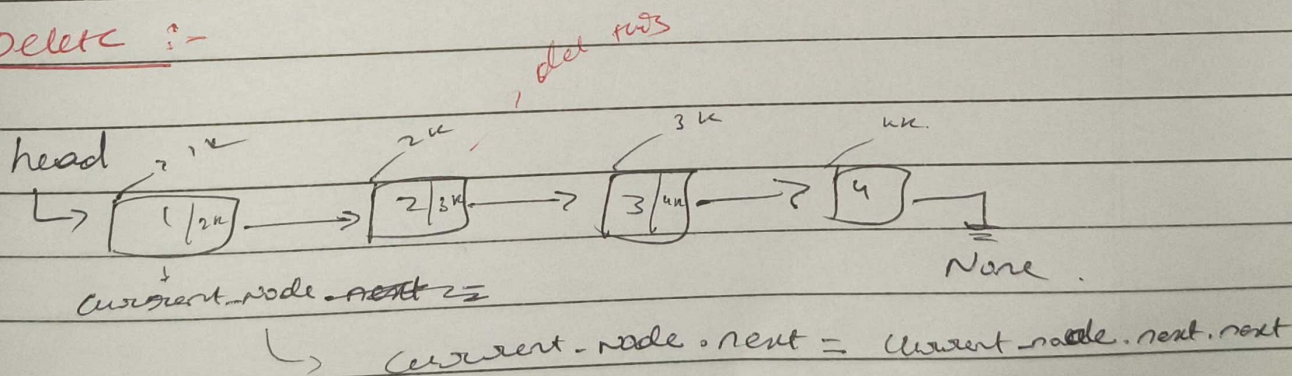


$$\text{Position } 3 = 3 - 1 \Rightarrow (2)$$

① $\text{new-node.next} = \text{current-node.next}$

② $\text{current-node.next} = \text{new-node}$

* Delete :-



```
14 class SinglyLinkedList:
15
16
17
18 def insert_at_position(self, data, insert_position):
19     # Position is invalid
20     if (insert_position <= 0):
21         print("Invalid position")
22         return
23
24     # Insert at the first position
25     if (insert_position == 1):
26         self.insert_at_beginning(data)
27         return
28
29     # Insert at positions 2 or greater
30     current_node = self.head
31     current_position = 1
32
33     while current_position < insert_position-1 and current_node is not None:
34         current_position = current_position + 1
35         current_node = current_node.next
36
37     if (current_node == None):
38         print("Invalid position, there are lesser number of nodes")
39         return
40
41     new_node = Node(data)
42     new_node.next = current_node.next #1
43     current_node.next = new_node #2
44
```

```
def delete_at_position(self, delete_position):
    # Position is invalid
    if (delete_position <= 0):
        print("Invalid position")
        return

    # Insert at the first position
    if (delete_position == 1):
        self.delete_at_beginning()
        return

    # Insert at positions 2 or greater
    current_node = self.head
    current_position = 1

    while current_position < delete_position-1 and current_node is not None:
        current_position = current_position + 1
        current_node = current_node.next

    if (current_node == None):
        print("Invalid position, there are lesser number of nodes")
        return

    current_node.next = current_node.next.next
```

```
def insert_delete_at_given_position(list: SinglyLinkedList):  
    list.insert_at_position(10, -1)  
    list.delete_at_position(-1)  
  
    list.insert_at_position(10, 1)  
    list.delete_at_position(1)  
  
    list.insert_at_end(10)  
    list.insert_at_end(20)  
    list.insert_at_end(30)  
  
    list.insert_at_position(15, 2)  
    list.insert_at_position(40, 4)  
    list.insert_at_position(100, 10)
```

I

```
if __name__ == "__main__":  
    # Create a new singly linked list  
    list = SinglyLinkedList()  
  
    # insert_at_beginning_driver_code(list)  
  
    # insert_at_end_driver_code(list)  
  
    # print_search_list_driver_code(list)  
  
    # delete_operations_start_end(list)  
  
    insert_delete_at_given_position(list)
```